

Module 1: Introduction to Cyber Security – One-Page Note

1. What is Cyber Security?

Cyber Security refers to the practice of protecting computers, networks, software, and data from unauthorized access, damage, or attacks. It ensures the confidentiality, integrity, and availability of information. As digital systems continue to expand, cyber security has become essential for individuals, organizations, and governments to stay safe online.

2. Importance of Cyber Security

- Protects sensitive data from theft and misuse
- Prevents financial losses due to fraud or hacking
- Helps maintain trust in online services
- Ensures business continuity by reducing downtime
- Protects national security and critical infrastructure

Cyber security is crucial because cyber attacks are increasing in frequency and sophistication, targeting both personal users and large organizations.

3. Common Types of Cyber Attacks

• Malware

Malicious software such as viruses, worms, and ransomware designed to damage or gain control over systems.

• Phishing

Fraudulent messages (often emails) used to trick users into revealing personal information or passwords.

• Denial of Service (DoS)

Attacks that flood a network or system to make it unavailable to users.

• Man-in-the-Middle Attack (MITM)

Attackers secretly intercept communication between two parties.

• Password Attacks

Attempts to gain unauthorized access by cracking or guessing passwords.

4. Cyber Security Principles

- **Confidentiality**

Ensuring only authorized people can access sensitive information.

- **Integrity**

Ensuring data is accurate and not tampered with.

- **Availability**

Ensuring systems and data are available when needed.

5. Cyber Hygiene Practices

Simple habits users can follow to stay secure:

- Use strong and unique passwords
 - Enable two-factor authentication
 - Avoid clicking unknown links or attachments
 - Keep software updated
 - Use antivirus and firewall protection
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6. Risk Management Basics

Cyber security also involves identifying risks, evaluating their impact, and implementing strategies to mitigate them. Organizations conduct risk assessments to detect vulnerabilities and strengthen their defense mechanisms.

7. Conclusion

Introduction to Cyber Security focuses on understanding the digital threat landscape and learning the foundational principles required to protect data and systems. Mastering these basics is essential before moving into advanced topics such as network security, cryptography, and ethical hacking.